

12.5 Writing Descriptions of Real-World Contexts From Algebraic Expressions

Lesson Introduction

	Benchmark	MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.			Learning Target	I can write a description of a real-world context from an algebraic expression.
	Benchmark Coherence	Building On MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions.			Working Toward MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
	Essential Questions	- How can an algebraic expression represent a real-world situation? - How do you create a real-world situation to describe an algebraic expression? - Can an algebraic expression describe more than one real-world situation?				
	Lesson Narrative	This lesson challenges students to make sense of algebraic expressions by creating real-world situations that represent the expressions. Students will have the opportunity to work with their partner to determine if any errors have been made and identify similarities between real-world situations. Students will practice composing expressions to represent real-world situations. Students need to read each story carefully to formulate an expression to represent it.				
	Component Summary	12.5.1 Warm-Up (~5 min.)	Math Story: Madam C.J. Walker (SE p. 304)	12.5.3 Collaboration (10-15 min.)	Create real-world contexts based on algebraic expressions (SE p. 305)	
		12.5.2 Activity (10 min.)	Match real-world contexts to expressions (SE p. 304)	12.5.4 Activity (15 min.)	Participate in a gallery walk to create more real-world contexts based on expressions (SE p. 306)	
		12.5.5 Wrap-Up (~5 min.)	Explain the difference between two expressions (SE p. 306)			
	Resources	Glossary		Materials		Additional Preparation
		<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Algebraic expression - Real-world scenario		(Optional) Crayons (Optional) Gallery Walk for 12.5.4		- During the second half of the 12.5.3 Collaboration, collect the scenarios that students did not use. Write those scenarios on chart paper and hang them around the room for the 12.5.4 Activity. - Another option is to create your own scenarios before class and hang them on the walls while students work on the second half of the 12.5.3 Collaboration.

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may struggle to differentiate between multiplication and division or addition and subtraction. Students who only look for keywords but do not thoroughly read the intent of the question may only use a key word like “per” or “give” and assume multiplication and addition even though both those keywords could mean any operation.	- Before beginning the lesson, consider briefly brainstorming with students some situations in which they use math in their daily lives. List general ideas on the board so students can refer to them during the lesson. - The MA.6.AR.1.1 benchmark is extended to having students write instead of just translating. This is an example of taking something that is of moderate complexity and extending it to a higher complexity. By extending student thinking to applications the student’s mind makes stronger connections to the contexts and their algebraic representations. Be prepared with scaffolding for students who may struggle but encourage students to persevere through using a growth mindset. - Consider doing small-group intervention with students who have been struggling while other students are doing the gallery walk.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display While pairs are working, circulate, collect, and make a visual display of vocabulary, phrases, and representations students use as they solve each situation. Make connections between how similar ideas might be communicated and represented in different ways. This helps students use mathematical language during paired and whole-group discussions.	MA.K12.MTR.7.1: Real-World Contexts Students will be working to create real-world scenarios that an algebraic expression could represent. Clarifications within this standard emphasize connecting mathematical concepts to everyday experiences. Use the lesson to provide opportunities for students to create models and compare their models with those created by other students.
 Differentiate Instruction	Consider having students preview the On-Ramp video titled Write and Interpret Expressions, or to have the video available for students to watch throughout the lesson.	
Lesson Notes:		

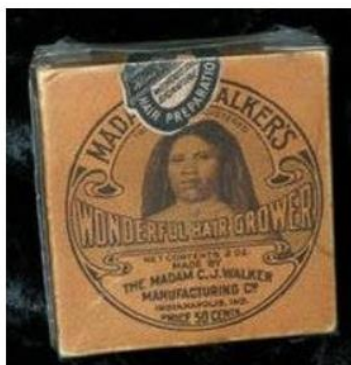
12.5.1 Warm-Up: Self-Made Millionaire

Component Overview: Students are introduced to Madam C.J. Walker and write their own algebraic expressions that could represent her revenue.



12.5.1 Warm-Up: Self-Made Millionaire

Madam C.J. Walker was the first African American self-made millionaire. She made her fortune by inventing and selling specialized hair products, including an innovative Edge Control product that sold for \$0.35 each beginning in 1905.



1. Create an algebraic expression to describe the total revenue Madam C.J. Walker earned for selling any number of Edge Control products, e .

$0.35e$



Consider an extension project for students to complete to accompany their work throughout the unit.

- For each lesson, students must provide and explain one example from that lesson that would fit a context for her edge-control business.
- Partner with an English or social studies teacher and complete a biography on her life and her resilience in her story.
- Watch a clip from “Self-Made” and facilitate a class discussion on the clip chosen.



Students could produce a variety of answers, so prompt them to define their variable and explain the constants they have included in the equation.

12.5.2 Activity: Matching Real-World Contexts to Expressions

Component Overview: Students match the relationship of a real-world situation to an algebraic expression that represents it. Remind students that they practiced this skill in the previous lesson.



12.5.2 Activity: Matching Real-World Contexts to Expressions

- Match each real-world situation with the correct expression.

A. Ryan has 20 songs. His friend bought him x more songs. How many songs does Ryan have?

B. Karen has x flowers. Helen has 20 times more flowers than Karen. How many flowers does Helen have?

C. George earned 20 points in a card game. Bruno earned x fewer points than George. How many points did Bruno earn?

D. Maria has 20 books and wants to put an equal number of books on x bookshelves. How many books go on each shelf?

1. $20x$

2. $x + 20$

3. $\frac{20}{x}$

4. $20 - x$



Consider having early finishers create a visual representation of one or more of the situations/expressions. Encourage students to think of ways to creatively convey the situation(s) as an image.



For each situation/expression pair, discuss the following questions as time allows:

- How did you figure out which expression matches the scenario?
- How did you determine the operation described in the scenario?

12.5.3 Collaboration: Writing Real-World Contexts

Component Overview: Students work with a partner or small group to compose a real-world scenario for each algebraic expression. Remind students that this is the opposite of what they did in the previous lesson when they wrote an algebraic expression to represent a real-world situation. After students select their four scenarios and while they are completing the second table, collect the scenarios they did not use. These scenarios can be used for the 12.5.4 Activity.



12.5.3 Collaboration: Writing Real-World Contexts

- For each expression, create a real-world scenario that the algebraic expression could represent.

Expression	Your Real-World Scenario	What does the variable represent?
$r - 10$	Answers will vary. Ana has r number of ribbons. She gives 10 to a friend. How many ribbons does Ana have left?	the number of ribbons
$21y$	Answers will vary. Zara is making bracelets that contain 21 beads per bracelet. She is making y bracelets for several friends. How many beads will she need?	the number of bracelets she is making
$\frac{100}{a}$	Answers will vary. The class has 100 pieces of dried fruit that they are equally splitting into a bags. How many pieces of fruit are in each bag?	the number of bags
$17 + s$	Answers will vary. Sean has 17 toy cars. For his birthday, he gets s more cars. How many cars does Sean have in all?	the number of cars Sean gets for his birthday
$57 - r$	Answers will vary. Mrs. Ruth has 57 thumbtacks. She gives r thumbtacks to another teacher. How many thumbtacks does she have left?	the number of thumbtacks Mrs. Ruth gives to another teacher
$48w$	Answers will vary. There are 48 pencils in a pack. Mr. Suarez bought w packs of pencils. How many pencils does he have?	the number of packs of pencils Mr. Suarez bought



If students struggle with math vocabulary, then creating a real-world scenario will be difficult. Consider providing students with a vocabulary reference that is visible during the class period. Include words such as “variable, difference, product, sum, splitting equally,” and “quotient.”



Before students begin, it may be helpful to discuss what it means for a variable to represent something. Remind students that they are working with algebraic expressions, not trying to solve a problem for a mathematical answer. Point out that in some situations, the variable may represent a whole amount, while in other situations, the variable may represent part of the problem. If students struggle to begin, consider working through the first one or two scenarios as a whole group.

12.5.3 Collaboration: Writing Real-World Contexts (continued)

2. With your partner, pick four expressions from Question 1 and use them to complete the following table.

Expression	Your Partner's Scenario	Does the Expression Accurately Represent the Scenario?	Similarities Between Your and Your Partner's Scenarios for the Expression
<i>Answers will vary. Students should use the scenarios they created in the previous problem.</i>	<i>Answers will vary. Students should use the scenarios they created in the previous problem.</i>	☐ Yes ☐ No	<i>Answers will vary. Students should use the scenarios they created in the previous problem.</i>
		☐ Yes ☐ No	
		☐ Yes ☐ No	
		☐ Yes ☐ No	



Encourage students to look for similarities and differences in the scenarios. Consider having students use the following questions to discuss the scenarios with their partner:

- Do both scenarios accurately represent the algebraic expression?
- How are the scenarios different?
- How are the scenarios similar?

12.5.4 Activity: Gallery Walk

Component Overview: Students can rotate through the Gallery Walk individually, in pairs, or in small groups. In this Activity, students write algebraic expressions to represent real-world situations. Before students begin, tell them which way to rotate around the room and how and when you will indicate when it is time to move to the next scenario. Either use the unused scenarios students created earlier and/or use the scenarios on the optional black line master. Number each scenario or place a label on each so students can record when expressions go with which scenario on their paper. Students should not be expected to write the scenario on their paper but have the scenario number and its corresponding expression on their chart.



12.5.4 Activity: Gallery Walk

Several scenarios are displayed. For each scenario, write an algebraic expression to match.

Scenario	Expression
Answers will vary. Scenarios will be the ones students wrote in Problem 1 but did not use in Problem 2, or other scenarios that you choose to create.	



Use the following questions in student pairs or as a whole group to discuss the algebraic expressions that students wrote during the Gallery Walk:

- Why does that algebraic expression represent the scenario?
- How did you decide which operation to use?
- How did you decide where to place the variable and the numerical value?

12.5.5 Wrap-Up: Ticket Out the Door

Component Overview: In the Wrap-Up, students will describe the difference in the position of the minuend and the subtrahend in a subtraction expression. Remind students that they are explaining how and why the expression is constructed, not finding a solution for a mathematical equation or problem.



12.5.5 Wrap-Up: Ticket Out the Door

1. Frank has a plant that measured h inches tall last week. This week the plant is 8 inches tall.

- a. Circle the expression that represents the number of inches, h , the plant grew in a week.

$$8 - h$$

$$h - 8$$

- b. Explain your choice.

To determine how many inches the plant grew last week, we start with how tall the plant is this week, 8 inches, and then subtract the number of inches, h , the plant grew last week.

- c. For the expression not chosen, describe a real-world situation that the expression might represent.

Answers will vary. Frank made h hacky sacks to sell at the school carnival. On the first day, he sold 8 hacky sacks. How many hacky sacks does Frank have left to sell?



For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.



Interpreting algebraic expressions from a context has been scaffolded over the last several lessons. Use the ticket out the door to inform student understanding as well as student growth for these lessons. Consider writing students positive notes about their progress.